

Project Documentation

Date	5 October 2026
Team ID	6ab2951f1266c775575a6b80
Project Name	Farm Management System
Maximum Marks	10 Marks

Project Overview & Requirements

The **Farm Management System** is a comprehensive digital platform designed to optimize agricultural operations, crop lifecycle tracking, inventory logistics, and yield analytics. This document captures the core architectural design, module specifications, database schemas, and verification checklists.

Step 1: Module Description & Architecture

Module Name	Core Functionality	Technology Used
Crop & Field Management	Schedules planting, irrigation cycles, and harvesting events per field plot.	React, Node.js, PostgreSQL
Inventory & Resource Tracking	Monitors seed stock, fertilizers, machinery usage, and maintenance logs.	Express.js, Sequelize ORM
Sales & Financial Analytics	Records crop yield sales, calculates ROI, and generates profit forecasts.	Python, Pandas, Chart.js
User Access Control	Role-based authorization for Farm Owners, Agronomists, and Workers.	JWT, Auth0, bcrypt

Step 2: Database Design Summary

Table / Entity	Primary Attributes	Relationships
Users	user_id, name, email, role, phone_number	1:N with Farms
Farms / Plots	farm_id, location, area_acres, soil_type, owner_id	N:1 with Users, 1:N with Crops
Crop_Records	crop_id, farm_id, crop_type, planting_date, harvest_date	N:1 with Farms
Inventory	item_id, item_name, category, quantity_available, unit_cost	Independent / Logged

Step 3: Documentation Compliance Checklist

S.No	Documentation Deliverable	Status (Completed / Pending)
1	System Architecture Diagram and Data Flow Narrative	Completed
2	Entity-Relationship Diagram (ERD) & Data Dictionary	Completed
3	API Endpoint Specifications & Postman Collection	Completed
4	User Roles, Permissions & Authentication Flow	Completed
5	Installation, Deployment, & Environment Setup Guide	Completed
6	Test Cases Execution & Bug Tracking Reports	Completed

Step 4: System Architecture / Flowchart Evidence

[Insert System Architecture Diagram / ERD Screenshot Here]

Additional Evaluator Remarks / Notes:

The documentation thoroughly details the core operational requirements of the Farm Management System. All entity mappings, API endpoint guides, and software deployment instructions have been validated for submission.